

WAEC 2026 CHEMISTRY PRACTICAL PRACTICE MANUAL

1. TITRATION

A is a solution of 0.200mol dm^{-3} of Tetraoxosulphate(vi) acid (H_2SO_4)

B. Is a solution containing 4.0g of Sodium hydroxide, per dm^{-3} of solution.

a. Put A into the burette and titrate it against 20.0cm^3 or 25.0cm^3 portion of B, using methyl orange as indicator. Repeat the titration to obtain consistent titre values. Tabulate your results and calculate the average volume of Acid used.

b. From the results and information provided, calculate the

(i) Concentration of B in mol dm^{-3}

(ii) Concentration of B in g dm^{-3}

(iii) Concentration of A in g dm^{-3}

(iv)

The equation of reaction $\text{H}_2\text{SO}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$

(Na = 23, H =1, O =16)

TABLE OF VALUES

Volume of Burette used = $50.00\text{cm}^3 = 0.05\text{dm}^3$

Volume of Pipette used = $25.00\text{cm}^3 = 0.025\text{dm}^3$

Indicator used = methyl orange 🍊

Burette Reading (Cm^3)	Rough Reading	First Reading	Second Reading	Third Reading
Final Reading (Cm^3)	15.60	15.40	15.30	
Initial Reading (Cm^3)	0.00	0.00	0.00	
Volume of Acid used.(Cm^3)	15.60	15.40	15.40	

To determine the average volume of acid used:

$$A.V = 1^{\text{st}} \text{ titre} + 2^{\text{nd}} \text{ titre} / 2$$

$$A.V = 15.40 + 15.30 / 2$$

$$A.v = 15.35 \text{ cm}^3$$

To determine the concentration of A: will be treated in class!!!

Question 2 - SALT ANALYSIS

2. C is a mixture of two substances which are inorganic and organic, carry out the following experiments on C. Record your observations and identify any gas evolved. State the conclusion you draw from the result of each test.

1ai. Add 5cm³ of distilled water to a portion of C in a test tube, Stir thoroughly and heat with dilute Nitric acid and filter. Divide the resulting solution into 4 portions.

1aii. To the first portion of the solution from (ai) above, add sodium hydroxide in drops and then in excess.

1aiii. To the second portion from (ai) above add aqueous ammonia in drops and then in excess.

1aiv. To the third portion of the solution from (ai), add dilute hydrochloric acid.

1av. To the residue add 2cm³ of Fehling's solution A or B and then heat gently for 5 minutes.

1avi. Add some few drops of iodine solution to residue and record your observation.

TEST	OBSERVATION	INFERENCE
1ai. C + distilled water + heat + dil. HNO ₃	Chalky white solution is formed, with the evolution of a colourless gas which turns like water milky	CO ₂ gas present CO ₃ ²⁻ present
1aii. Filtrate of solution C + NaOH in drops then in excess	White precipitate formed Precipitate dissolves	Al ³⁺ , Pb ²⁺ , Zn ²⁺ suspected
1aiii. Filtrate of Solution C + NH ₃ in drops Then in excess	White precipitate formed White precipitate still persist	Al ³⁺ , Zn ²⁺ Pb ²⁺ suspected Al ³⁺ and Pb ²⁺ suspected
1aiv. Filtrate of solution C + Dil. HCl	White chalky precipitate formed, with no gas evolved. Precipitate does not dissolved	Pb ²⁺ confirmed

1av. Residue of mixture of C + Fehling's solution A and B + heating

1avii. Residue of mixture C + drops of iodine solution

No colour change observed and residue remain undissolved

Blue Black colouration observed

Reducing sugars absent

Starch is present